

A blue parallelogram and a light green parallelogram are positioned in the upper-left corner of the slide. The blue shape is partially behind the green one. Both shapes are oriented diagonally, with their longer sides running from the top-left towards the bottom-right. The background of the entire slide is a dark navy blue with subtle, lighter blue diagonal stripes running from the bottom-left towards the top-right.

The Blank OSL

Presented by: Team Vertex (Alan, Evan, Pouria, Shirley, Tony)



Team Vertex

- Team of 5 members
 - All on campus this term and next term
 - Diverse expertise and background
 - Shared vision of building AI-driven application
- Organization
 - Weekly standup
 - Github -> Gitlab
 - Discord



Advisor / Mentor

Ramani T.

Research Engineer

@ Google DeepMind



Recap of timeline



Pitch Presentation

Research MCP
Setup Dev Env
Brainstorm MCP Servers

Midterm Presentation

Upgrade to MCP Servers
User Confirmation Feature
Setup Claude Desktop

Final Presentation

Support Various Platforms
Continue Updating/Adding
Servers



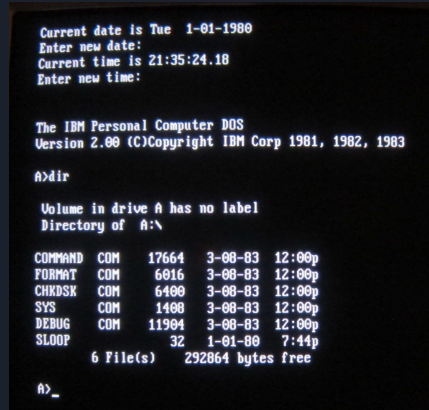
Recap: Blank Operating System Layer

- **LLM-Powered Intuition**
 - Interprets natural language for expressive OS control.
- **Secure, Modular Design**
 - Uses specialized servers and the Model Context Protocol (MCP).
- **Execution Requires Approval**
 - Tasks are always reviewed and confirmed by the user.

Evolution of User Interface

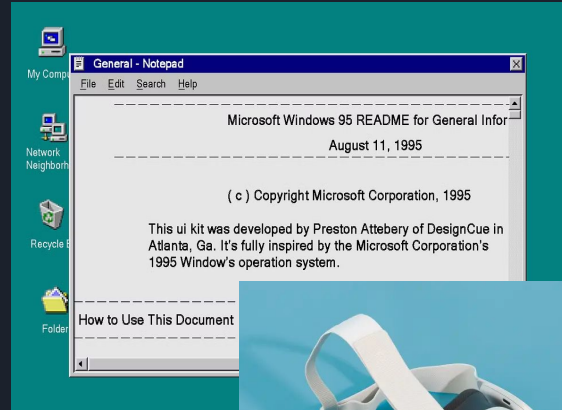
1970s - 1980s

Text-based user interface



1980s - 2007s

Graphical user interface

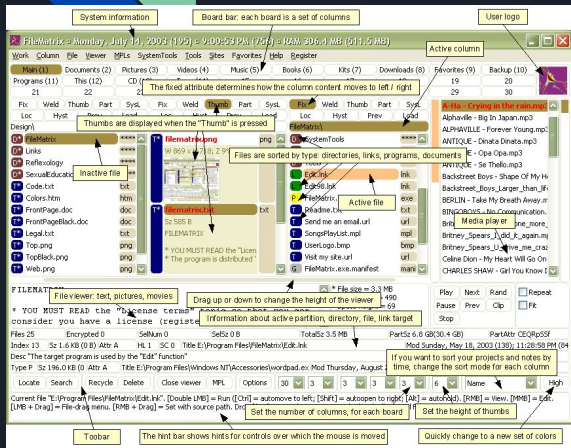


2007s - Present

Finger/Voice-based user interface



Our vision



Stupidly simple.



Stupidly intuitive.

How can you make money as a kid?

Google:



Bing:

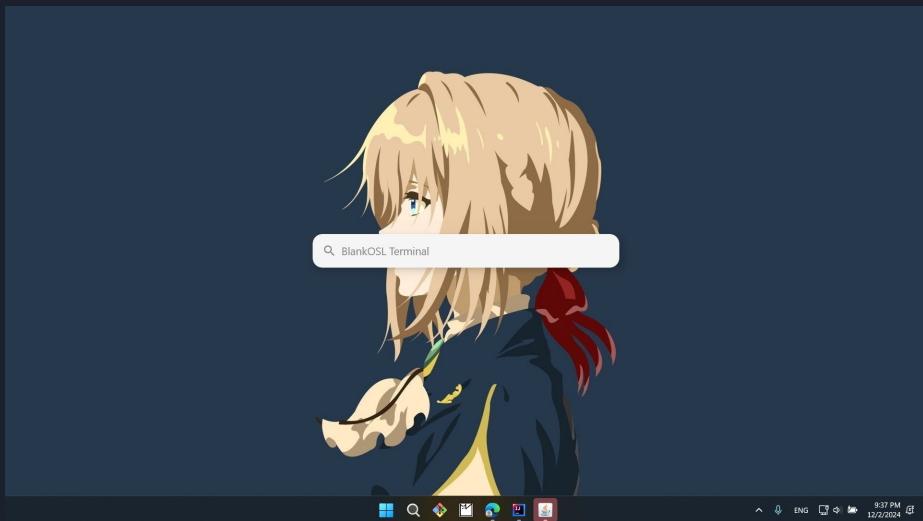
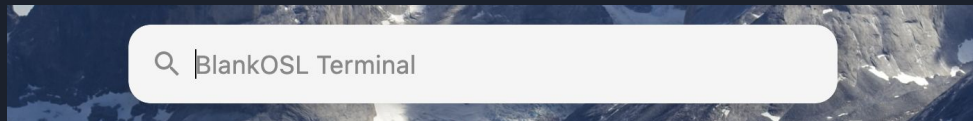


Stupidly smart.

Project Overview

The Blank OSL:

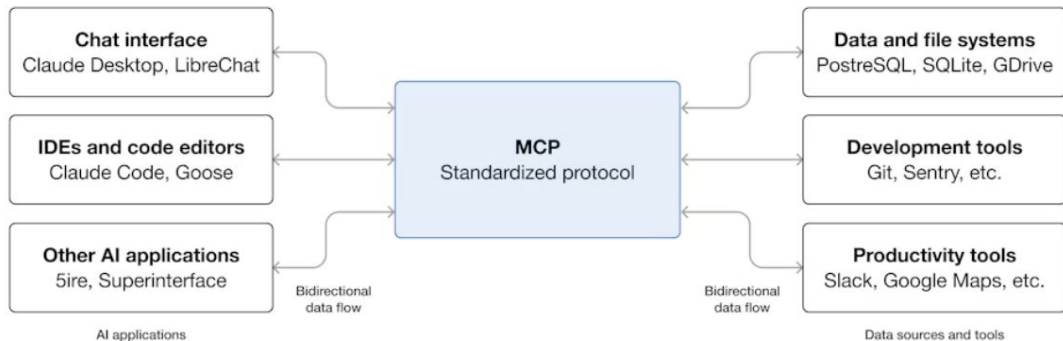
An Operating System Layer (OSL) utilizes LLMs to improve OS navigation.



- The UI of all UI.
- Minimalist UI design.
- Beginner friendly with minimum knowledge required.
- Fast-paced environment that requires quick solution.

Pivot to Model Context Protocol (MCP)

Think of MCP like a **USB-C** port for AI applications. MCP provides a standardized way to connect AI applications to external systems.





Roadmap

The Model Context Protocol is rapidly evolving. This page outlines our priorities for **the next release on November 25th, 2025**, with a release candidate available on November 11th, 2025.

Statelessness and Scalability

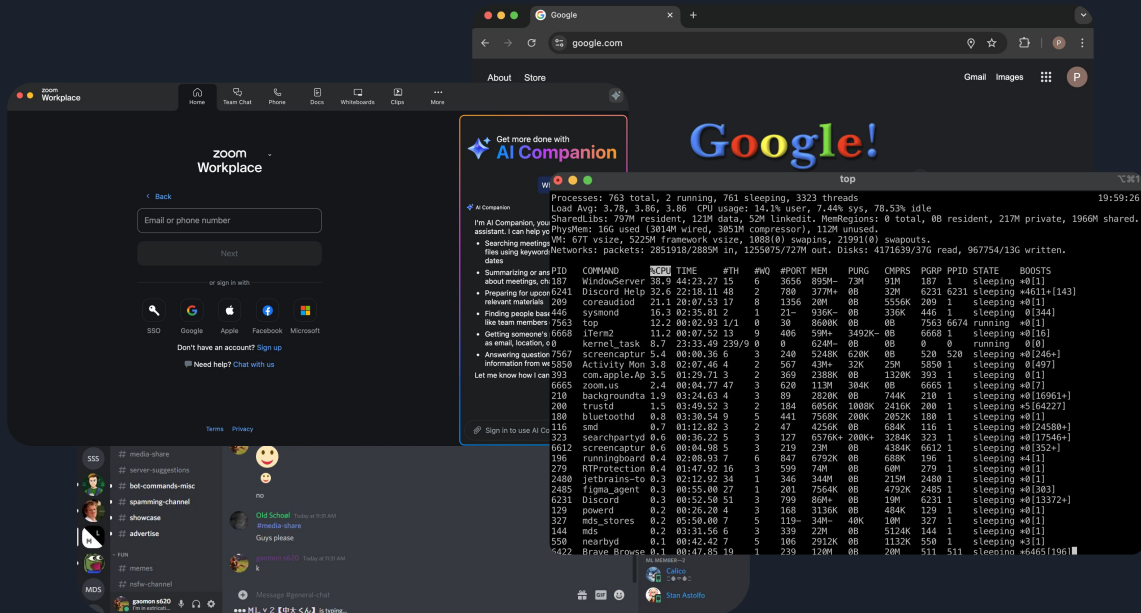
Asynchronous Operations MCP Registry General Availability

SDK Support Standardization Server Identity

Official Extensions **Validation**

Automate Systems

- OS File System
- Task Manager
- Sending emails
- PDF
- Dedicated HTTP Clients
 - Weather
 - Stock
 - ...



[Blog](#) / [Product](#) / [Article](#)

PDFs to Production: Announcing state-of-the-art document intelligence on Databricks

With `ai_parse_document`, parse and understand PDFs directly in SQL with leading quality at 3-5x lower cost



Published: November 11, 2025

[Product](#)

5 min read

By [The Mosaic Research Team](#)

Summary

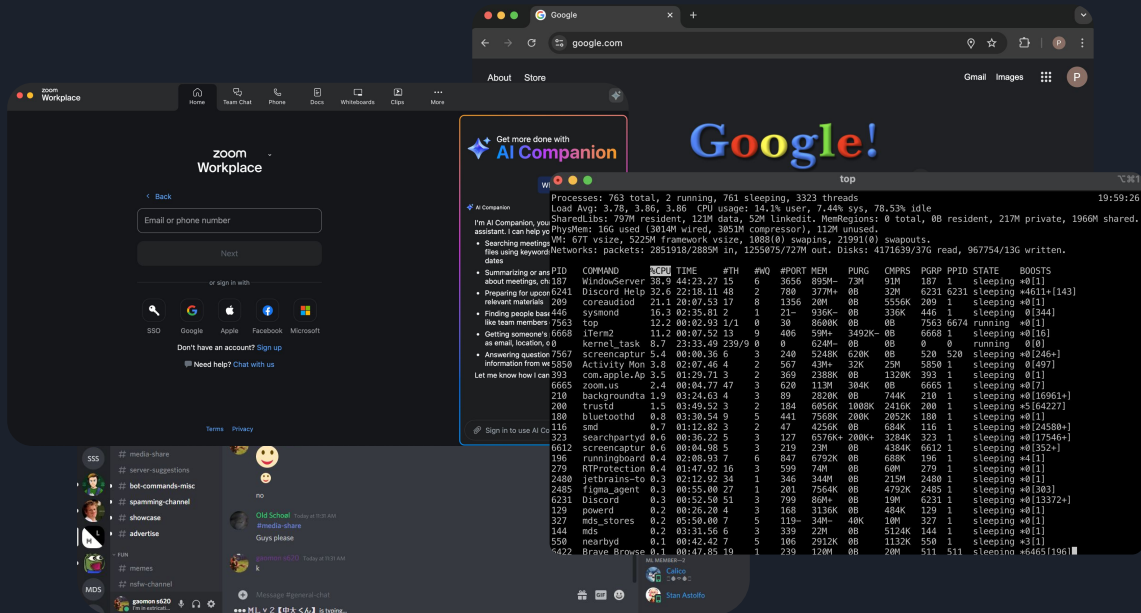
- Today, we're introducing the newest addition to Agent Bricks: `ai_parse_document` (Public Preview): Powered by our research team's agentic system for large-scale multimodal

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Automate Systems

- OS File System
- Task Manager
- Sending emails
- PDF
- Dedicated HTTP Clients
 - Weather
 - Stock
 - ...





Implementation Challenges

- **Filesystem**
 - Sandbox Security / Unauthorized Write/Delete Files
 - Directory Traversal
- **Email**
 - Credential Management / Third-Party API Access (Gmail)
- **PDF**
 - Complex Content Parsing (e.g., SVG, Charts)
 - Conversion (HTML → PDF)
- **Other MCP Server**
 - API Key Exposure / Sensitive Data Leakage
 - Rate Limit Failure / Accidental Throttling

Live Demo with Claude!





Command #1

Hi! I am Miss Fang, a math teacher and would like to prepare a PDF worksheet with 10 addition problems.

Can you create the PDF for me and save it in a folder called **Math Practice** in my desktop folder?



Command # 2

Great! Can you help me email my student with the above question sheet attach?

Please email Shirley y266fang@uwaterloo.ca to ask her to finish the questions by this week.

Can you also email Shirley's parent fangyixuan0123@gmail.com with a pdf of answers attached to it?

Current and Future Progress

- Upgrade to MCP ✓
- User Confirmation Feature ✓
- Various Feature Improvements (end of term)
- Support for Different Platforms (end of term)
- Finding testing users (end of term)
- Speech to Text Feature (TBD)

BlankOSL Terminal

s Claude wants to use Get stock price from stocks

Request

```
{  
  "symbol": "GOOGL"  
}
```

Deny Esc

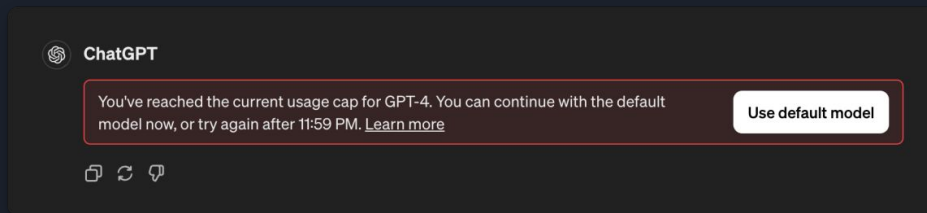
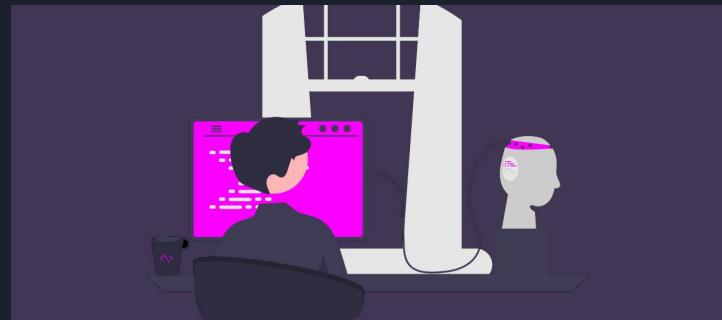
Always allow % ↵

Allow once ↵



Challenges

- Running out of tokens/prompts
 - Claude Pro
- High Response Latency
 - Switching to faster models
- Developer environment (Mac vs Win)





Thank you!

Questions?